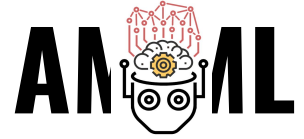




**Politecnico
di Torino**



Advanced Machine Learning

A.A. 2025/2026

Tatiana Tommasi

Basic Information

Teacher: Tatiana Tommasi

<http://www.tatianatommasi.com/>

<https://scholar.google.com/citations?user=ykFtl-QAAAAJ&hl=en>
tatiana.tommasi@polito.it

Support from:

Paolo Rabino, Leonardo Iurada, Claudia Cuttano, Davide Sferrazza...

Prof. Barbara Caputo, Giuseppe Averta, Carlo Masone, Raffaello Camoriano, Francesca Pistilli

Visual and Multimodal Applied Learning Laboratory (Vandal Lab)

<http://vandal.polito.it/>

<https://ellis.eu/>



e l l i s

European Laboratory for Learning and Intelligent Systems

Click on the dots for more information about each unit.

On the Map **List of units**

Alicante	Amsterdam	Barcelona	Berlin
Cambridge	Copenhagen	Darmstadt	Delft
Edinburgh	Lausanne (EPFL)	Zürich (ETH)	Freiburg
Genoa	Graz	Haifa (Technion)	Heidelberg
Helsinki	Jena	Vienna (IST Austria)	Leuven
Linz	Lisbon	Lviv	London (UCL)
Madrid	Manchester	Milan	Modena (Unimore)
Munich	Nijmegen	Oxford	Paris
Potsdam	Prague	Saarbrücken	Stuttgart
Tel Aviv	Trento	Tübingen	Turin
Sofia	Warsaw		

Unit Director



Tatiana Tommasi
Scholar

Unit Members



Giuseppe Averta
Member



Raffaello Camoriano
Member



Barbara Caputo
Fellow



Stefano Favaro
Member



Giulia Fracastoro
Member



Enrico Magli
Fellow



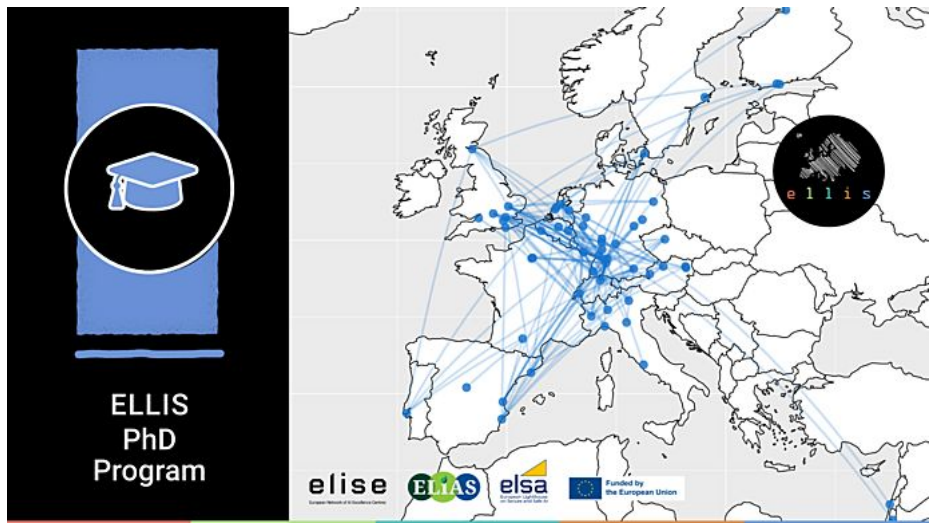
Carlo Masone
Member



Giuseppe Rizzo
Member



Diego Valsesia
Member



<https://ellis.eu/news/ellis-phd-program-attracts-global-ai-talent-to-europe-120-early-career-researchers-join-the-network-in-2024>

<https://ellis.eu/websites/european-lab-for-learning-intelligent-systems/pages/what-it-s-like-to-be-an-ellis-phd-student>

<https://ellis.eu/phd-postdoc>

<https://ellis.eu/news/ellis-phd-program-call-for-applications-2025>

COLLABORATE

with two leading researchers from labs across Europe in machine learning related research!

FOCUS

on curiosity-driven research and excellence in machine learning and related areas

PARTICIPATE

in boot camps, summer schools and workshops, and build your network

Basic Information

Code: 01URWOV, 01URWYG

Master of science in Computer Engineering (Ingegneria Informatica)

Credits: 6CFU

When/Where: Tuesday 16:00-19:00, room 12A
Friday 11:30-13:00, room 3P

How: in presence + online (rare pre-recorded videos) + discussion sessions

Exercises: in presence

everything will be recorded with exception of the exercise sessions.

Discord channel:

<https://discord.gg/rAzc7Eb8dr>

It will be useful mainly to interact with the project mentors



**Politecnico
di Torino**



A short survey

Do you know what is Principal Component Analysis (PCA)?

Do you know what is Support Vector Machines (SVM)?

Have you ever trained an SVM classification model?

Do you know what is cross validation?

Do you have experience with python coding?

Have you ever trained a CNN classification model?



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A short survey

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Do you know what is cross validation?

Do you have experience with python coding?

Have you ever trained a CNN classification model?

Do you know what the acronym GPT stands for?

Useful Material

Deep Learning. Ian Goodfellow, Yoshua Bengio, Aaron Courville, <https://www.deeplearningbook.org/>

The little book of Deep Learning <https://fleuret.org/francois/lbdl.html>

Deep Learning, Foundations and Concepts, Bishop <https://link.springer.com/book/10.1007/978-3-031-45468-4>

Foundations of Computer Vision, <https://visionbook.mit.edu/>

Artificial Intelligence A Modern Approach. Stuart J. Russel and Peter Norvig

An Introduction to Probability Theory and its applications. William Feller

Machine Learning: a probabilistic perspective. K. P. Murphy.

Understanding Machine Learning: From Theory to Algorithms. Shai Shalev-Shwartz and Shai Ben-David

CVPR

NeurIPS

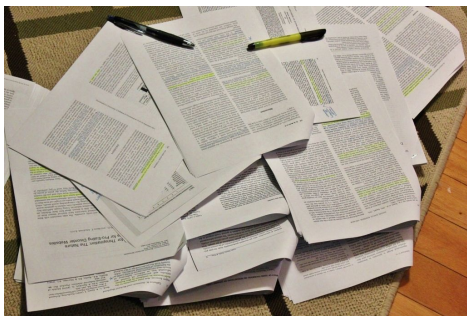
ECCV

ICCV

ICLR

...

arXiv



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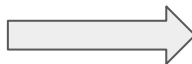
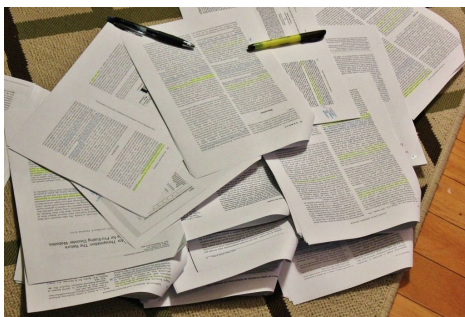
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online - Convolutional Neural Networks for Visual Recognition - Fei Fei Li - <http://cs231n.stanford.edu/>

online - Machine Learning course on Coursera - Andrew Ng - <https://www.coursera.org/>

polito - Data Spaces - 01RLPOV, 01RLPNG, Prof. Vaccarino

polito - Machine Learning and Pattern Recognition - 01URTOV, Prof. Cumani

Pre-requirement: Python basic elements

Online tutorial for Python: <https://docs.python.org/3/tutorial/>

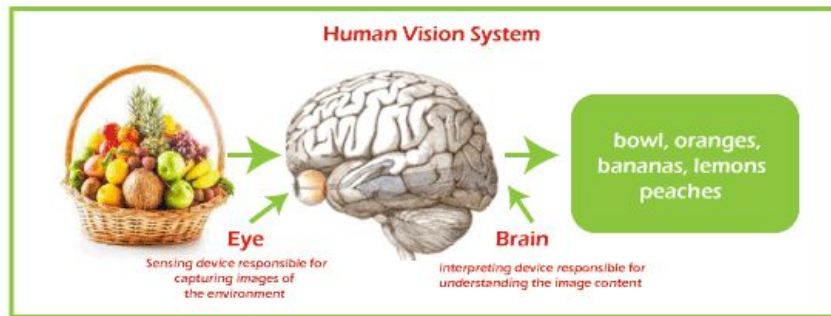
Online tutorial for Pytorch: <https://pytorch.org/tutorials/>

Topics and Objective

- AI, ML basic definitions
- probability fundamentals and an elementary probability-driven classifier
- loss, risk, prediction rules
- ~~shallow supervised/unsupervised learning~~
- deep learning (supervised, unsupervised, recurrent, generative)

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- ~~shallow supervised/unsupervised learning~~
- deep learning (supervised, unsupervised, recurrent, generative)
- **computer vision applications**

+ 3d geometry

+ natural language processing

Course Aim

- You will get to know the basics of ML & DL
- You will learn how to read & understand a paper (code included)
- You will learn how to design & implement your own ML/DL algorithm for a given problem
- You will learn how to explain what you know about ML/DL to others

Course Aim

- You will get to know the basics of ML & DL

~~mix of in-person, online live and recorded lectures~~

- You will learn how to read & understand a paper (code included)

individual/small group interactive work guided by me and by teaching assistants

- You will learn how to design & implement your own ML/DL algorithm for a given problem

work in groups mainly guided by the teaching assistants

- You will learn how to explain what you know about ML/DL to others

presentation in front of the class / at the exam

Exam Modality

Paper Presentation + Project + Oral Exam

Exam Modality

Paper Presentation + Project + Oral Exam

By mid-November (it might vary a bit) we will have covered basics DL

Then the lectures will be dedicated to presentation of seminal papers of the last 3-7 years (roughly) on various DL topics. The topics will be of interest for all projects.

You will present papers in groups: **4 students, 1 paper**

Papers will be assigned by me in random order.

You will know the title of your paper about 2 weeks before your presentation. There will be some time to adjust the calendar.

You will have **12 minutes sharp** for the presentation.

You will be judged on your comprehension of the papers and on your ability to explain it.

Minimum individual score: 6

Maximum individual score : 10

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HOW

There will be a lecture/video dedicate to how to present a paper.

WHY????

Because working with ML/DL means being **a life-long-learner!**

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HOW

There will be a lecture/video dedicate to how to present a paper.

WHY????

Because working with ML/DL means being **a life-long-learner!**

Paper presentation is very instructive but difficult to organize with 320 registered students who might not all attend in presence.

Thus, this year the paper presentation is not mandatory.

You might decide to not register to present the paper → it implies a **different approach at the oral (◡)**

Exam Modality

Paper Presentation + Project + Oral Exam

At the beginning of November one lecture/video will be dedicated to possible project research-areas.

There will be >4 proposals, for each proposal there will be some mandatory work plus suggestions on how to extend it.

You will work in teams of 4 people.

Project Proposal Submission: Before Christmas break (more details on date / submission modality will come), you will indicate: (1) your team (2) the project you chose

Final report due: ten days before the oral exam

The report will be a pdf of max 8 pages using [a provided template](#)

You can keep your project till the last exam session before the next course beginning (1 year)

Minimum group score: 6

Maximum group score : 10

Exam Modality

Paper Presentation + Project + Oral Exam

You will make a group presentation (about 15 mins) of your project and a Q&A session will follow

The goal is to evaluate both your project work and your individual **basic course knowledge(*)**([○]) .

Minimum individual score : 6

Maximum individual score : 10

Important Note (1) : This is not a trial-error exam

Important Note (2): the code of your project will be checked - don't cheat

(*) mandatory: this usually consists of one or two questions covering all the course topics discussed in class. Insufficient answers to those questions imply failing the exam.

([○]) not presenting the paper will imply more mandatory course topic questions at the oral

Questions: Paper Presentation

Paper Presentation:

- Can I chose the paper?
No
- I cannot make the presentation on my assigned date because ...
You can ask a colleague for a switch: in this case I have to receive an email/discord message in advance explaining why the switch happens and who will present the paper. No impact on the score either way.
- Will I get guidance/help in preparing the presentation?
Yes there will be one lecture or at least I will share a dedicate video about how to read & how to present a scientific paper/work. You will not get individual help or support to try out the presentation.

Questions: Project

- Can I propose my own project?
No
- Can I do the project alone/in two?
No. If you don't know anyone we'll match you
- What computing facilities I can use for the project?
Google Colab
- How will I chose the project?
There will be >4 projects, each led by a different teaching assistant.
In November each project will be presented with a lecture & live Q&A to help you choosing
- I am not proficient in python, will I be able to make the project?
It will be tough but you can exploit a lot of online material to learn python (if you want to make a living in ML you better learn it!)
- Will I get guidance/help in working on the project?
Yes you will be supported by the teaching assistants both with meetings and on discord

Questions: Oral

- Can I make the oral exam in a different session than the rest of my group?
No
- Is the oral exam in English?
Yes
- What will be asked during the exam?
Everything related to the project, from implementation details to theory related to the algorithms used in the project. There will be also mandatory questions on any topic explained during the lectures.
- Why are we doing this project???? 
This is the main question I'm going to ask you during the exam...
- Can I reject the final vote and sit again the exam?
Yes. For the next exam session you will need to search for another team on a different topic (i.e. you will have to work on a different project). Please come to sit at the exam when you are prepared.

Why Are you attending this course?

Learn about basic tools that you will need for:

- **your M. Sc./PhD Thesis**
- **your Start Up**
- **Wall Street**
- **Google, Facebook/Meta, Amazon, Snapchat, ...**
- **Huawei, Samsung, IBM, Siemens, Microsoft, Apple,...**
- **Disney, Netflix...**
- **Linkedin, AirBnB, Booking,...**
- **Uber, BMW, Honda, Toyota, Tesla, FCA, iRobot, ...**

MACHINE INTELLIGENCE 3.0

ENTERPRISE INTELLIGENCE

VISUAL

Orbital Insight planet
clarifai DEEP VISION
cortica Ilogicon
SPACE KNOW Captricity
netra deepomatic

AUDIO

Gridspace TalkiO
nexidia twilio
CAPIO Expect Labs
Clover Mobvoi
QuirousAI popUP archive

SENSOR

PREDIX GIoT MAANA
Sentenai PLANET OS
UPTAKE IMUBIT Preferred Networks
thingworx KONUX Alluvium

INTERNAL DATA

PRIMER IBM WATSON
Gycorp Palantir ARIMO
Alation Sapho Outlier
Digital Reasoning

MARKET

mattermark Quid
Datafox PREMISE
Bottlenose MOTIVA
enigma CB INSIGHTS
Tracxn predata

CUSTOMER SUPPORT

DigitalGenius Kasisto
Eloquent WISEIO
ACTIONIQ zendesk
Preact CLARABRIDGE

SALES

collective[i] osense
fuse|machines AVISO
salesforce INSIDE SALES
Zenight clari

MARKETING

MINTIGO Lattice RADIUS
LiftIgniter PERSADO
brightfunnel retention
COGNICOR AIRPR msgai

SECURITY

CYLANCE DARKTRACE
ZIMPERIUM deepinstinct
Sentinel DEMISTO
graphistry drawbridge
SignalSense AppZen

RECRUITING

textio entelo
Wade & Wendy hiQ
unifive SpringRole
GIGSTER HireVue

AUTONOMOUS SYSTEMS

GROUND NAVIGATION

drive.ai AdasWorks
ZOOX MOBILEYE
UBER Google TESLA
nuro Autonomy Aero Robotics

AERIAL

SKYDIO SHIELD AI
Airware DJI LILY
DroneDeploy
pilotAI SKYCATCH

INDUSTRIAL

JAYBRIDGE OSARO
CLEARPATH fetch
KINDRED rethink robotics
HARVEST

AGENTS

PERSONAL

amazon alexa
Cortana Allo
facebook
Siri Replika

PROFESSIONAL

butter.ai pogo SKIPFLAG
clara x.ai slack
talla Zoom sudo

INDUSTRIES

AGRICULTURE

BLUERIVER mavrx
tile TRACE Pivotal Bio
TerraVision AGRI-DATA
Descartes Labs udie abundant

EDUCATION

KNEWTON volley
gradescope
CTI coursera
UDACITY alt school

INVESTMENT

Bloomberg sentient
ISENTIUM KENSHO
alphasense Dataminr
CEREBELLUM CAPITAL Quandl

LEGAL

blue J BEAGLE
Everlaw RAVEL
Sseal ROSS
LEGAL ROBOT

LOGISTICS

NAUTO Acerta
PRETECKT
Routific clearmetal
MARBLE PITSTOP

INDUSTRIES CONT'D

MATERIALS

zymergen Citrine
Eigen Innovations
SIGHT MACHINE
SINKBO nanotronics
BIOWORKS CALCULARIO

RETAIL FINANCE

TALA zest finance
Lendo earnest
affirm MIRADOR
wealthfront Betterment

HEALTHCARE

PATIENT

PULSE CareSkore
ZEPHYRUS HEALTH IBM Watson Health
Oncoda SENTRIAN
Atomwise Numerate

IMAGE

BUTTERFLY 3SCAN
ARTERYS enlitic
BAYLABS imagia
Google DeepMind

BIOLOGICAL

iCarbonX color GRAIL
deep genomics RECURSION
LUMINIST Numerate
Atomwise verily WHOLE BIO

TECHNOLOGY STACK

AGENT ENABLERS

OCTANE.AI howdy. Maluuba KITT.AI
OpenAI Gym Kasisto AUTOMAT
semantic machines

DATA SCIENCE

DOMINO SPARKBEYOND rapidminer
kaggle DataRobot yhat AYASDI
data iku seldon yseop bigml

MACHINE LEARNING

CognitiveScale GoogleML context relevant
Cycorp HyperScience narda logics minds.ai H2O.ai
SCALED INFERENCE sparkcognition loop GEOMETRIC INTELLIGENCE
deepsense.io reactive skymind bonsai

NATURAL LANGUAGE

agolo FLYLIEN LEXALYTICS
Narrative Science loop@ spaCy LUMINOSO
cortical.io MonkeyLearn

DEVELOPMENT

SIGOPT HyperOpt fuzzyio okite
rainforest lobe Anodot
Signifai LAYER6 bonsai

DATA CAPTURE

CrowdFlower Diffbot CrowdAI import io
Paxata DATA SIFT amazon mechanical turk enigma
WorkFusion DATALOGUE TRIFACTA parsehub

OPEN SOURCE LIBRARIES

Keras Chainer CNTK TensorFlow Caffe
H2O DEEPLARNING4J theano torch
DSSTNE Scikit-learn AzureML neon
MXNet DMTK Spark PaddlePaddle WEKA

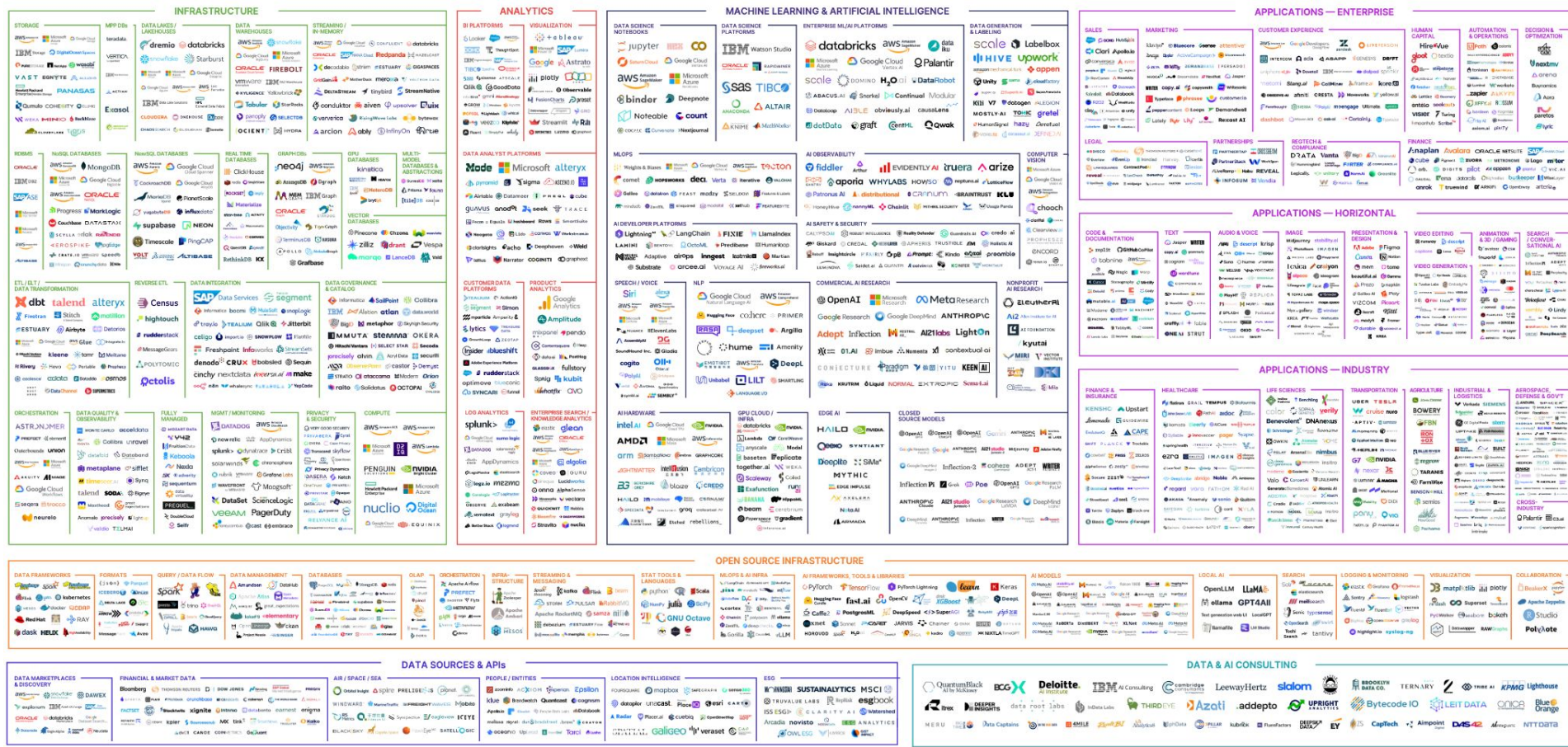
HARDWARE

KNUPATH TENSTORRENT Cirrascale
NVIDIA intel nervana Movidius
tensilica GoogleTPU 1026 Labs Qualcomm
Cerebras Isosemi

RESEARCH

OpenAI mnsense ELEMENT vicarious
KNOGGIN Numenta Kimera Systems Cogital

THE 2024 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE



Version 1.0 - March 2024 © Matt Turck (@mattturck), Aman Kabeer (@AmanKabeer1) & FirstMark (@firstmarkcap) Blog post: mattturck.com/MAD2024 Interactive version: MAD.firstmarkcap.com Comments? Email MAD2024@firstmarkcap.com



<https://mad.firstmarkcap.com/>

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Caveat...

- NOT: learn to use deep learning as a black box
- YES: understand models and architectures

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- **NOT:** learn to use deep learning as a black box
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This is not an easy course, but results speak for themselves!

Why Are you attending this course?

Top 10 jobs involving AI skills

Top jobs seeking machine learning or artificial intelligence skills

Rank	Job title	% of postings containing AI or machine learning	Rank	Job title	% of postings containing AI or machine learning
1.	Machine learning engineer	75.0%	6.	Algorithm developer	46.9%
2.	Deep learning engineer	60.9%	7.	Junior data scientist	45.7%
3.	Senior data scientist	58.1%	8.	Developer consultant	44.5%
4.	Computer vision engineer	55.2%	9.	Director of data science	41.5%
5.	Data scientist	52.1%	10.	Lead data scientist	32.7%

Source: Indeed



AI job titles with the highest salaries



Rank	Job title	Average salary
1.	Machine learning engineer	\$142,858.57
2.	Data scientist	\$126,927.41
3.	Computer vision engineer	\$126,399.81
4.	Data warehouse architect	\$126,008.25
5.	Algorithm engineer	\$109,313.51